



Dimensionality Reduction (Part 1 of 2): Linear Methods

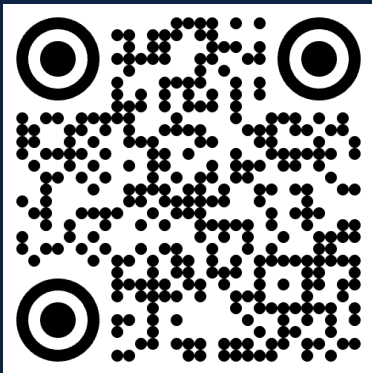
The Advanced R Series: Clustering & Classification

Slides and code created by Jesse Ghashti

March 23, 2026

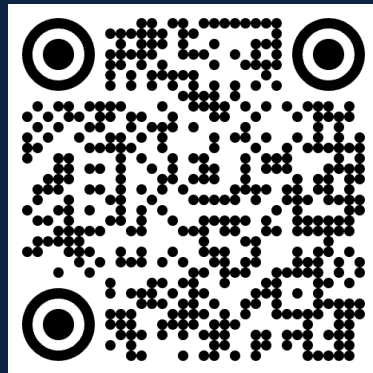
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←
Check out our other
CSC workshops!

GitHub code and
slides for today's
workshop (and pre-
vious workshops)



Alternatively, code/slides available at the bottom of
<https://csc-ubc-okanagan.github.io/workshops/>

Workshop Series Overview



Session	Topic	Date/Time
1	Hierarchical Models	Jan 14, 12:00 PM
2	Centroid-Based Models	Jan 19, 3:30 PM
3	Fuzzy Clustering	Jan 28, 1:00 PM
4	Distribution-Based Models	Feb 2, 3:30 PM
5	Density-Based Models	Feb 9, 3:30 PM
6	Graph-Based Models	Feb 23, 3:30 PM
7	Mixed-Type Data Quantification	Mar 4, 1:00 PM
8	Mixed-Type Data Clustering	Mar 11, 1:00 PM
9	Bias Reduction and Fairness	Mar 18, 1:00 PM
10	Dimensionality Reduction 1 of 2	Mar 23, 3:30 PM
11	Dimensionality Reduction 2 of 2	Mar 30, 3:30 PM

What we learned...

- Clustering can produce biased results even without using protected attributes
- Defined group fairness metrics: balance, disparate impact, composition deviation
- Defined individual fairness: similar individuals should be treated similarly
- Applied fairlets — decompose data into balanced pairs before clustering
- Applied linear residualization — remove protected attribute first moment dependencies from variables
- Observed the fairness–quality tradeoff on COMPAS recidivism data

Today -> Dimensionality Reduction



Today we will...

- Understand the curse of dimensionality and why it breaks clustering
- Learn Principal Component Analysis (PCA) for variance-based reduction
- Learn Factor Analysis (FA) for latent structure modeling
- Apply both methods to the UCI Communities and Crime dataset
- Compare clustering performance: full space vs reduced space
- Interpret principal components and factors

Today we require...

```
library('ggplot2')  
library('dplyr')  
library('tidyr')  
library('cluster')  
library('mclust')  
library('psych')
```

High-dimensional data creates problems for clustering

The Curse of Dimensionality

- As p (number of variables) grows, pairwise distances tend towards uniform
- All points appear “equally far apart” — no structure to exploit
- Distance-based methods (K -Means, PAM, Spectral) lose discriminative power
- Many features may be noise unrelated to the structure of interest

Dimensionality reduction addresses this by

- Finding a lower-dimensional representation
- Concentrating signal in fewer dimensions
- Filtering out noise

Two Approaches Today



Principal Component Analysis

Find orthogonal directions of maximum variance

- Data-driven transformation
- No probabilistic model
- Components ordered by variance
- Primarily for *data reduction*

Factor Analysis

Model observed variables as driven by latent factors

- Probabilistic generative model
- Assumes latent structure
- Factors can be rotated
- Primarily for *interpretation*

Both produce lower-dimensional representations useful for clustering. The choice depends on whether you need to *explain* the dimensions (as we will see)

Socioeconomic and Crime Data from US Communities

Data Sources

- 1990 US Census (socioeconomic variables)
- 1990 LEMAS survey (law enforcement)
- 1995 FBI UCR (crime statistics)

Structure:

- $n = 1994$ communities across the US
- $p = 98$ predictive features (after cleaning)
- Features normalized to $[0, 1]$
- Target: ViolentCrimesPerPop (violent crimes per capita)

Category	Example Variables
Demographics	racepctblack, racePctWhite, agePct12t29, agePct65up
Economics	medFamInc, perCapInc, PctUnemployed, PctPopUnderPov, etc...
Education	PctNotHSGrad, PctBSorMore, PctLess9thGrade, etc...
Family Structure	PctKids2Par, PctFam2Par, TotalPctDiv, PctIlleg, etc...
Housing	PctHousOwnOcc, MedRent, PctVacantBoarded, HousVacant, etc...
Immigration	PctForeignBorn, PctRecentImmig, PctSpeakEnglOnly, etc...

With 98 features, many are correlated. This suggests a lower-dimensional structure underlies the data — exactly what PCA and FA are designed to find.

Naive Labels -> Higher vs Lower Crime Communities



For evaluating clustering, we create a binary classification

- **Higher Crime** are communities with `ViolentCrimesPerPop` in the top 35th percentile
- **Lower Crime** are communities with `ViolentCrimesPerPop` in the lower 65th percentile

Note these are arbitrarily chosen labels for demonstrative purposes. We use these labels to *demonstrate* that dimensionality reduction improves clustering interpretation when signal is buried in noise.

Strongest Positive Correlations

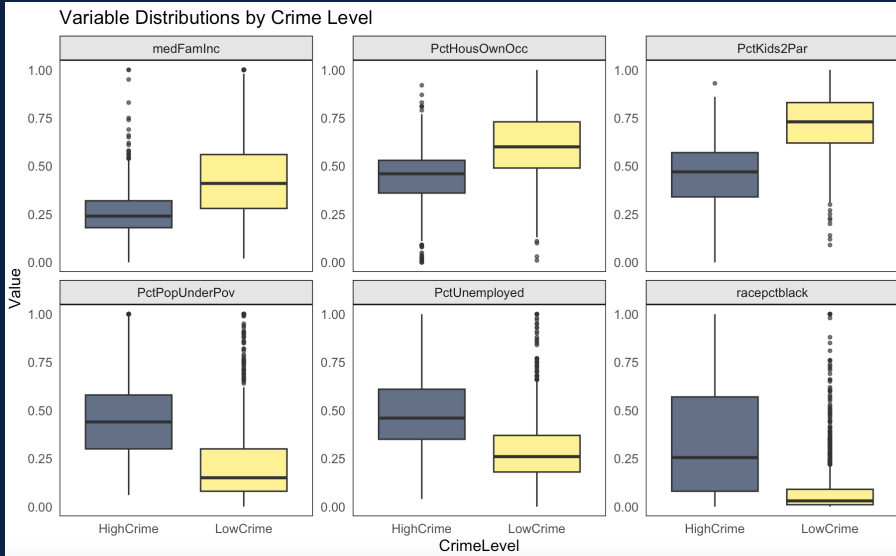
PctIlleg: +0.74
racepctblack: +0.63
racepctblack: +0.58
FemalePctDiv: +0.56
TotalPctDiv: +0.55
MalePctDiv: +0.53
PctPopUnderPov: +0.52
PctUnemployed: +0.50

Strongest Negative Correlations

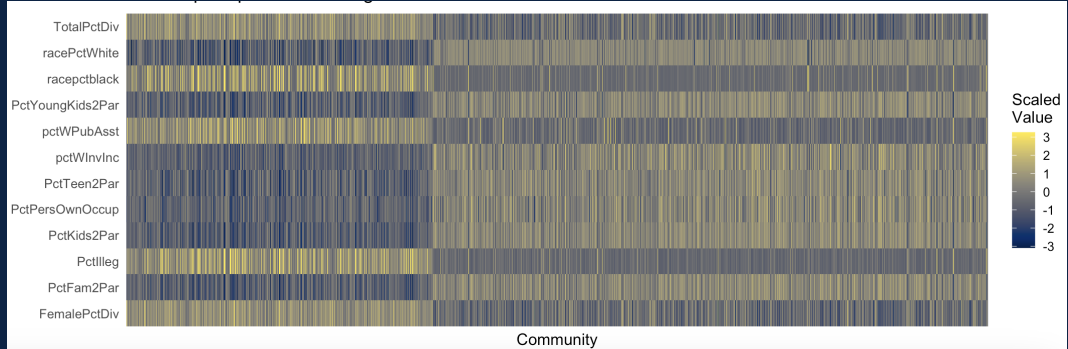
PctKids2Par: -0.74
PctFam2Par: -0.71
racePctWhite: -0.68
PctYoungKids2Par: -0.67
PctTeen2Par: -0.66
pctWInvInc: -0.58
PctPersOwnOccup: -0.53
PctHousOwnOcc: -0.47

Many features are strongly correlated with crime — and with each other. This suggests a lower-dimensional structure underlies the data.

Select Variable Distributions by Crime Level



Discriminating Variables of Crime Level



Racial composition variables (`racePctBlack`, `racePctWhite`, amongst others) are among the strongest predictors of crime rate. If we cluster on these features, clusters will likely encode racial structure.

Recall from Workshop 9

- Clusters can have disparate impact on protected groups
- Balance metric: $\min(r, b) / \max(r, b)$ within each cluster
- Consider Fairlets, residualization, etc.

Disclaimer: We will not apply fairness methods today — our focus is dimensionality reduction. However, keep in mind that components/factors may encode sensitive attributes. Applying fair clustering to reduced representations is a possible exploration.

Find new axes that capture maximum variance

Setup

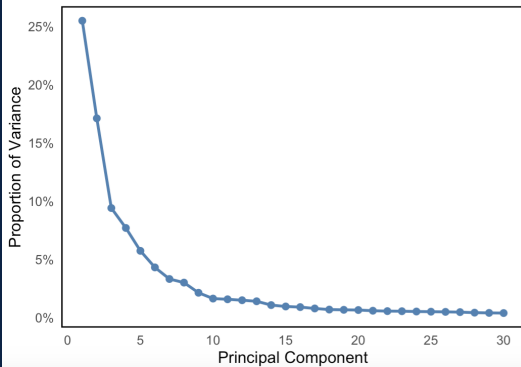
- Data matrix $\mathbf{X}$ ($n \times p$), centered and scaled
- Compute sample covariance $\mathbf{S} = \frac{1}{n-1} \mathbf{X}^\top \mathbf{X}$
- Eigendecomposition: $\mathbf{S} = \mathbf{V} \mathbf{\Lambda} \mathbf{V}^\top$
- Eigenvalues $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_p$ (variance per PC)
- Eigenvectors $\mathbf{v}_1, \mathbf{v}_2, \dots$ are the *loadings*

Properties

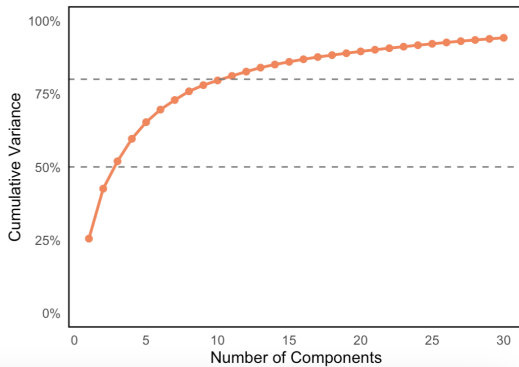
- PCs are orthogonal (uncorrelated)
- PC1 has maximum variance; PC2 has maximum variance orthogonal to PC1; etc.
- PC Scores: $\mathbf{Z} = \mathbf{XV}$ — coordinates in PC space

```
pcaResult <- prcomp(crimeFeatures, center = TRUE, scale. = TRUE)
eigenvalues <- pcaResult$sdev^2
propVar <- eigenvalues / sum(eigenvalues)
cumVar <- cumsum(propVar)
pcScores <- pcaResult$x
loadings <- pcaResult$rotation
```

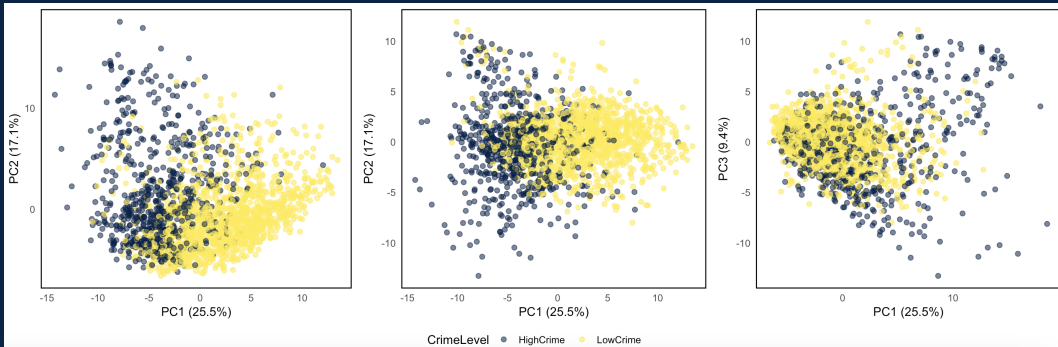
Scree Plot: Variance Per Component



Cumulative Variance Explained



PCA Projection -> Visualizing Structure



Clear separation between Higher and Lower crime communities emerges in just 2 dimensions.

Baseline -> Clustering in Full Space



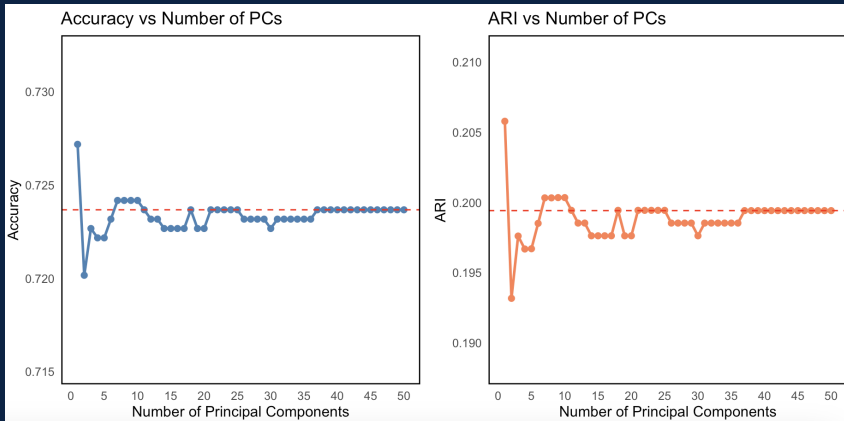
```
featureScaled <- scale(crimeFeatures)
set.seed(2026)
kmeansFull <- kmeans(featureScaled,
                      centers = 2,
                      nstart = 25)

ariFull <- adjustedRandIndex(
  kmeansFull$cluster, trueClass)
accFull <- max(
  mean(kmeansFull$cluster == as.numeric(trueClass)),
  mean(kmeansFull$cluster != as.numeric(trueClass)))
```

Full Space Clustering (98D)

Metric	Value
Accuracy	0.724
Adjusted Rand Index	0.199

Accuracy vs Number of PCs

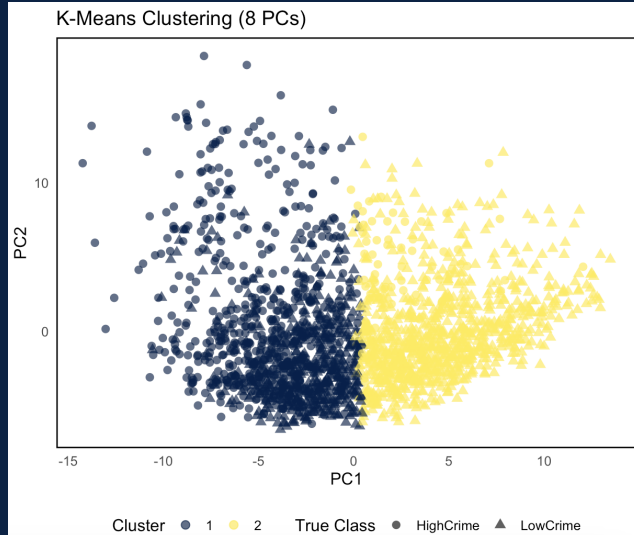


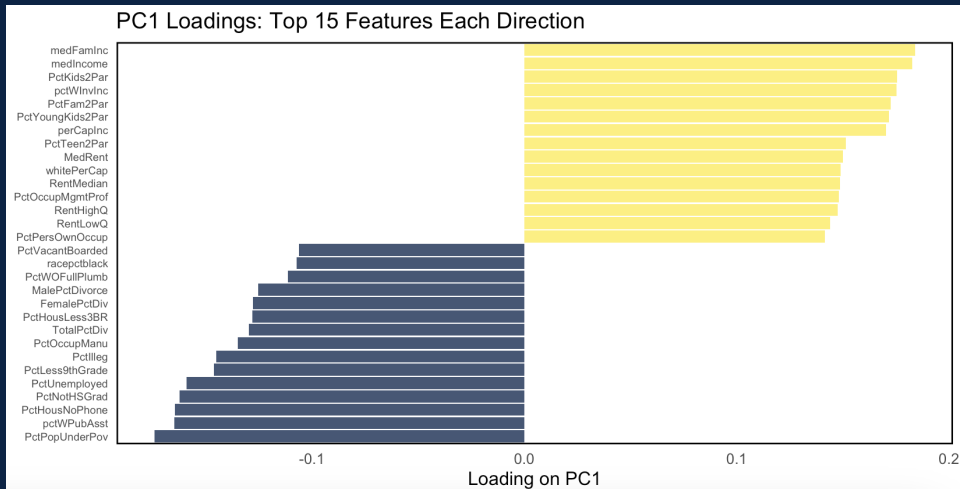
Interesting that we can achieve maximum accuracy with only 1 PC! However, we will use the second best performance (8 PCs) so we can analyze them further.

Method	Dimensions	Accuracy	ARI
Full Space	98	0.724	0.199
PCA (8 PCs)	8	0.724	0.200

Reducing from 98 to 8 dimensions reduces dimensions heavily while maintaining clustering quality. Concentrating signal in fewer dimensions helps K -Means find a better structure

Clustering Visualization in PC Space





What Does PC₁ Capture?



Negative Loadings (Low PC₁)

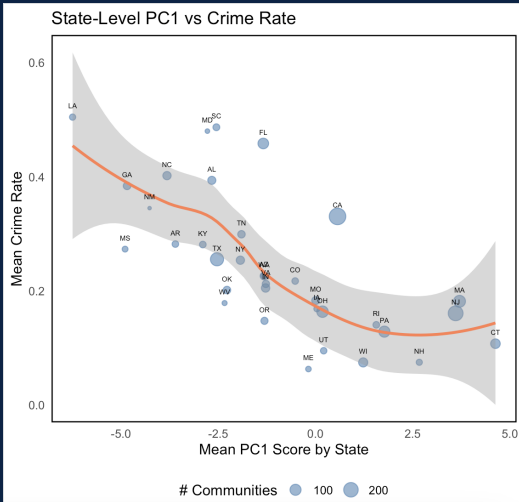
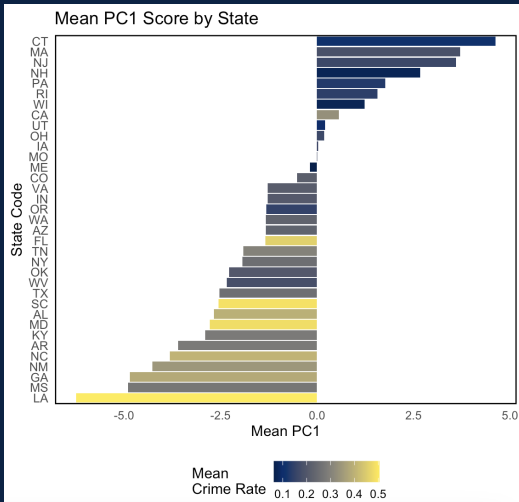
PctPopUnderPov
PctUnemployed
PctIlleg
racepctblack
PctHousNoPhone
etc...

Positive Loadings (High PC₁)

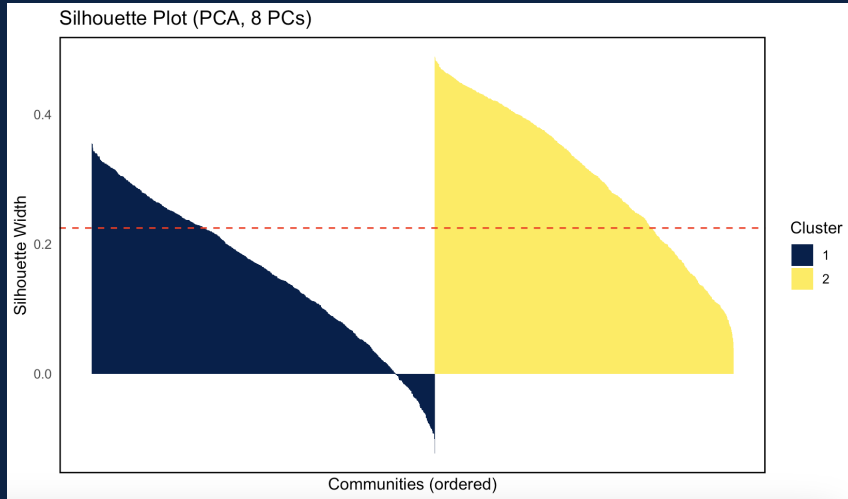
medFamInc
PctKids2Par
PctFam2Par
whitePerCap
pctWInvInc
etc...

PC₁ could be capturing a socioeconomic gradient: disadvantage (poverty, unemployment, single-parent families) vs advantage (income, home ownership, two-parent families).

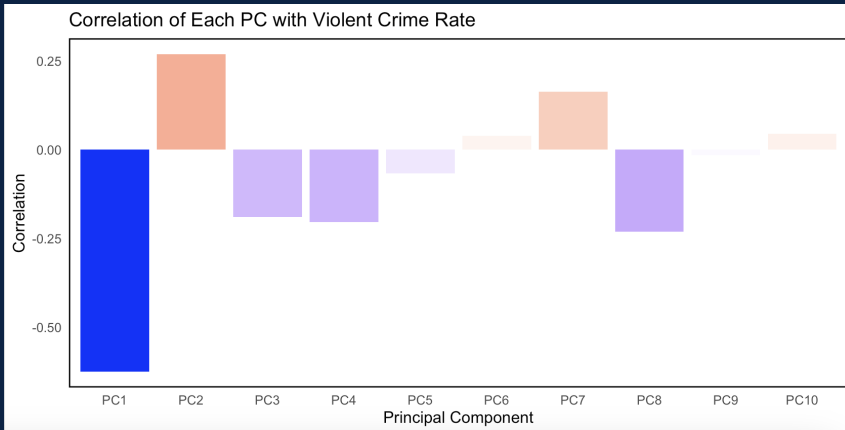
PC1 by State



Silhouette Scores



PC Correlations with Crime Rate



PC₁ has the strongest correlation with crime. Later PCs have near-zero correlation — they capture noise, not crime-related signal.

Observed variables are generated by latent factors

Model

$$\mathbf{x}_i = \boldsymbol{\mu} + \boldsymbol{\Lambda}\mathbf{f}_i + \boldsymbol{\epsilon}_i$$

- $\mathbf{x}_i$ = observed variables for observation i ($p \times 1$)
- $\mathbf{f}_i$ = latent factor scores ($k \times 1$, $k \ll p$)
- $\boldsymbol{\Lambda}$ = factor loadings matrix ($p \times k$)
- $\boldsymbol{\epsilon}_i$ = unique/error variance (diagonal covariance $\boldsymbol{\Psi}$)

Covariance decomposition

$$\text{Cov}(\mathbf{x}) = \boldsymbol{\Lambda}\boldsymbol{\Lambda}^\top + \boldsymbol{\Psi}$$

Common variance (shared across factors) + Unique variance (specific to each variable)

Aspect	PCA	Factor Analysis
Goal	Maximize variance	Model correlations
Model	No probabilistic model	Latent variable model
Variance	All variance is “common”	Common + unique variance
Components	Orthogonal by construction	Can rotate for interpretation
Primary use	Data reduction	Understanding structure
Solution	Unique	Rotation gives many solutions

FA can be used when you believe latent constructs generate the data and want interpretable factors.

Parallel Analysis

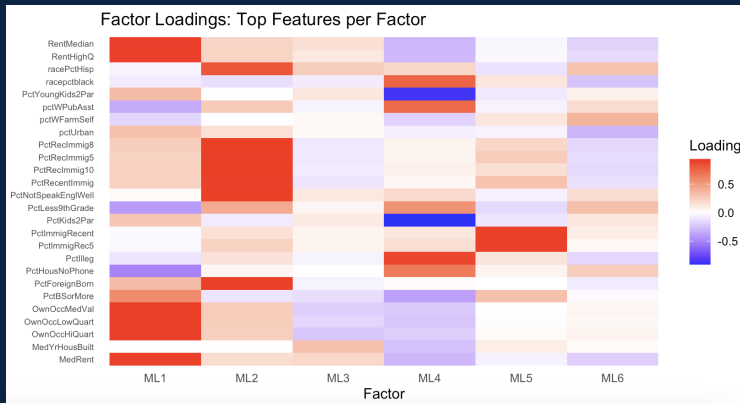
1. Generate random data with same n, p
2. Compute eigenvalues of random correlation matrix
3. Repeat; compute 95th percentile
4. Keep factors where real eigenvalue $>$ random

Other methods

- Kaiser: eigenvalue $>$ 1
- Scree test: look for elbow
- Interpretability

```
library(psych)
kmoResult <- KMO(featureSubset)
faParallel <- fa.parallel(
  featureSubset,
  fm = "ml",
  fa = "fa",
  n.iter = 20)
nFactors <- faParallel$nfact
faResult <- fa(featureSubset,
  nfactors = nFactors,
  rotate = "varimax",
  fm = "ml")
```

Factor Loadings Heatmap



Each row = a feature; each column = a factor. Red = positive loading; Blue = negative loading. Varimax rotation achieves “simple structure” — each feature loads highly on few factors.

Example Factor Interpretations

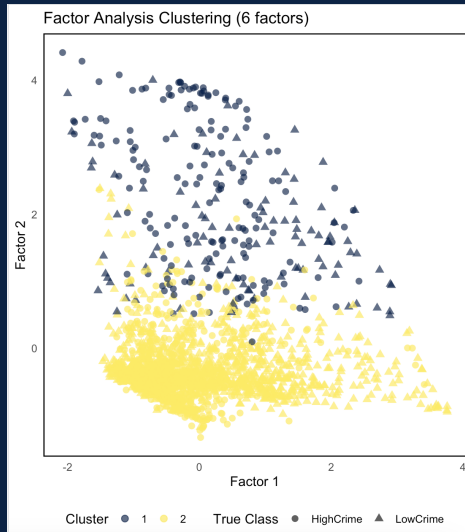
Factor	High Loadings	Interpretation
1	MedRent, OwnOccHiQuart, RentHighQ, ...	Housing
2	PctForeignBorn, PctRecentImmig8, ...	Immigration
4	PctIlleg, pctWPubAsst, PctHousNoPhone	(?)

Factors represent interpretable latent constructs underlying the data. Unlike PCA components, we can *name* them based on which variables load highly.

Clustering with Factor Scores



```
factorScores <- faResult$scores
set.seed(2026)
kmeansFA <- kmeans(factorScores,
                    centers = 2,
                    nstart = 25)
ariFA <- adjustedRandIndex(
  kmeansFA$cluster, trueClass)
accFA <- max(
  mean(kmeansFA$cluster == as.numeric(trueClass)),
  mean(kmeansFA$cluster != as.numeric(trueClass)))
```



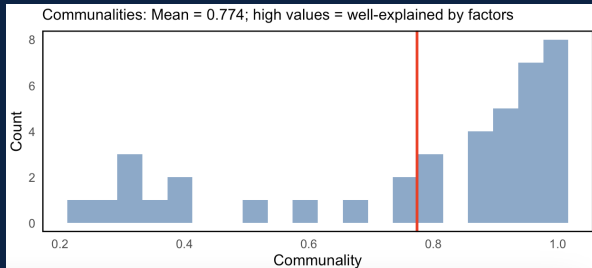
Communalities -> How Well Are Variables Explained?



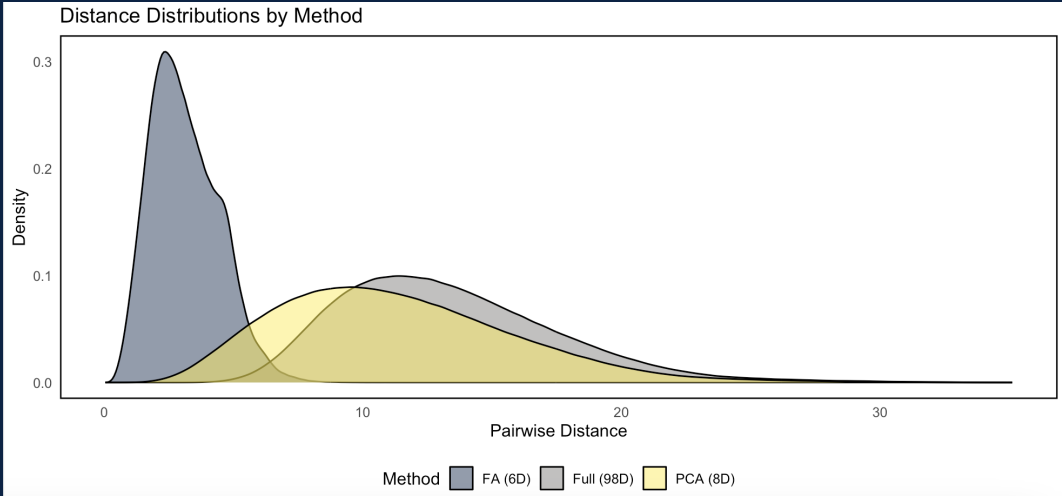
Communality (h^2) is the proportion of a variable's variance explained by the common factors. $h_j^2 = \sum_{k=1}^K \lambda_{jk}^2$, where λ_{jk} is the loading of variable j on factor k .

- $h^2 \approx 1$: variable is almost entirely explained by the factors
- $h^2 > 0.6$: well-explained by the factor model
- $h^2 < 0.4$: poorly explained — mostly unique variance

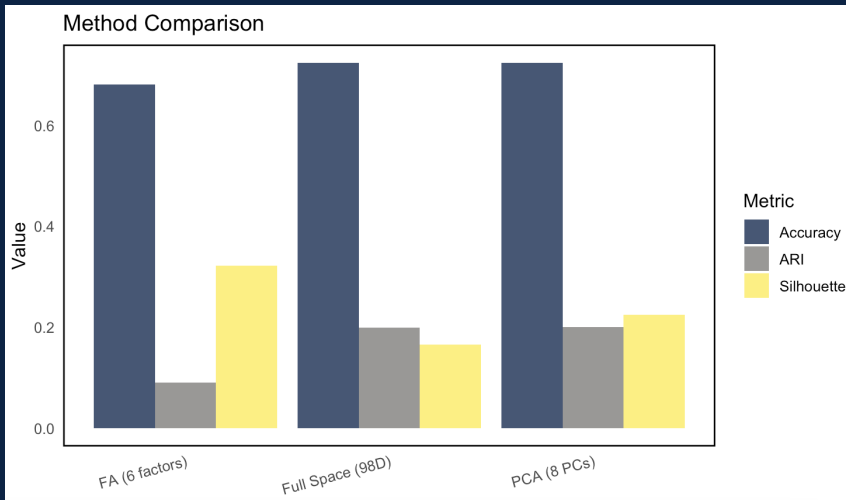
Feature	Communality
OwnOccMedVal	0.9967141
PctRecImmig8	0.9964132
RentMedian	0.9952356
PctRecImmig10	0.9933897
PctRecImmig5	0.9925707
OwnOccLowQuart	0.9899937
MedRent	0.9858167
OwnOccHiQuart	0.9823308
PctRecentImmig	0.9750282
RentHighQ	0.9729794



Distance Distributions -> Why Reduction Helps



Method Comparison



PC₁ is strongly correlated with racial composition

racecctblack: $r \approx -0.54$ with PC₁

racePctWhite: $r \approx 0.58$ with PC₁

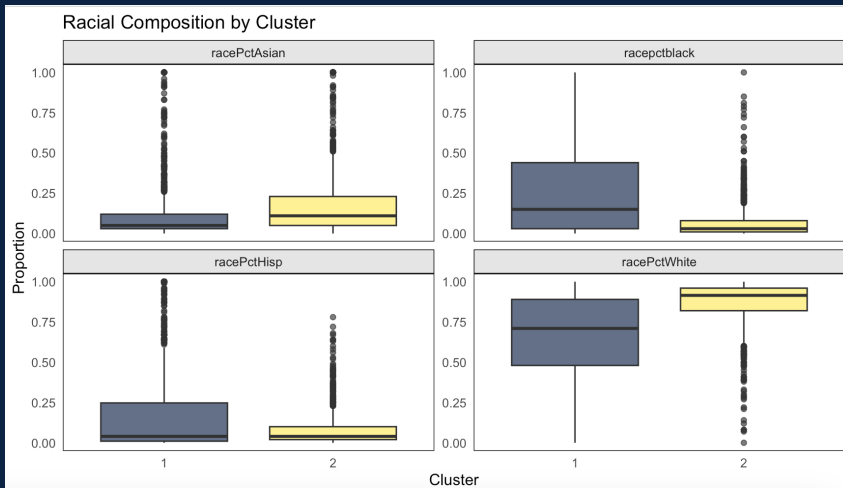
racePctAsian: $r \approx 0.24$ with PC₁

racePctHisp: $r \approx -0.39$ with PC₁

Exercise Idea

Does residualization preprocessing reduce this correlation? How does this change the structures of the Factors and PCs?

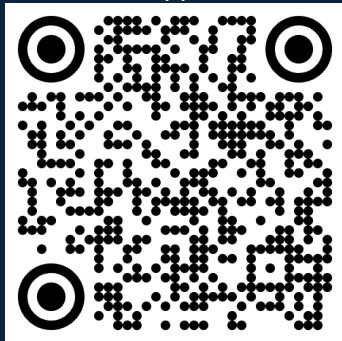
Racial Composition by Cluster



- High-dimensional data suffers from the curse of dimensionality: uniform distances, no structure
- PCA finds orthogonal directions of maximum variance; concentrates signal in first few components
- Factor Analysis models observed variables as driven by latent factors; provides interpretable dimensions
- Both methods substantially improve clustering over the full feature space
- Choosing the number of components/factors matters — more is not always better
- Interpretation is important: PC₁ captured the “socioeconomic gradient”
- Fairness concern: reduced dimensions may encode protected attributes (race)

Additional Questions?

Book an Appointment!



Next Workshop: Mar 30, 3:30 PM

Dimensionality Reduction 2 of 2

-> PCA and FA assume linear relationships.
What if structure is nonlinear?

-> Next workshop introduces t-SNE and UMAP
for nonlinear dimensionality reduction.

-> We'll compare linear vs nonlinear methods
and discuss when to use each.

Thank You!

Questions?

Workshop Materials:

<https://csc-ubc-okanagan.github.io/workshops/>

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